

name	income_group	pop_n	pop_u	wat_bas_n	wat_lim_n	wat_unimp_n	wat_sur_n	wat_bas_r	wat_lim_r	wat_unimp_r	wat_sur_r	wat_bas_n	wat_lim_n	wat_unimp_n	wat_sur_n	pop_n (m)	wat_bas_n (rounded)	wat_bas_n_clean	wat_lim_n_clean	wat_unimp_n_clean	wat_sur_n_clean	income_group_num
Tokelau		125000000	0	99.707665	0	0.292323459	0	99.707665	0	0.292323459	0	NAN	NAN	NAN	NAN	0	99.707665	99.707665	0	0.292323459	0	0
Niue		1618000031	46.2202348	97.0107618	0	2.989122822	0	97.0107618	0	2.989122822	0	NAN	NAN	NAN	NAN	0	97.0107618	97.0107618	0	2.989122822	0	0
Falkland Islands (Malvinas)	NAN	348300004	78.5079561	95.3089274	0	4.691072596	0	95.3089274	0	4.691072596	0	100	0	0	0	0	95.3089274	95.3089274	0	4.691072596	0	0
Montserrat	NAN	4996000072	9.14969771	98.07748262	0	1.922517378	0	98.07748262	0	1.922517378	0	NAN	NAN	NAN	NAN	0	98.07748262	98.07748262	0	1.922517378	0	0
Saint Pierre and Miquelon	NAN	5.793000078	89.96199799	91.8412	0	8.6	0	91.8412	0	8.6	0	NAN	NAN	NAN	NAN	0	91.8412	91.8412	0	8.6	0	0
Saint Helena	NAN	6.077000059	40.08200073	99.1	0.8	0	0	99.1	0.8	0	0	NAN	NAN	NAN	NAN	0	99.1	99.1	0	0.8	0	0
Saint Barthélemy	NAN	9.885	100	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Nauru	High income	10.83399693	100	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Tuvalu and Futuna Islands	NAN	11.24600029	0	99.14328736	0	0.856712639	0	99.14328736	0	0.856712639	0	NAN	NAN	NAN	NAN	0	99.14328736	99.14328736	0	0.856712639	0	0
Wallis	Upper middle income	17.9196982	64.0139994	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Cook Islands	NAN	17.83699118	75.49000075	99.97161022	0	0.0283987127	0	99.97161022	0	0.0283987127	0	NAN	NAN	NAN	NAN	0	99.97161022	99.97161022	0	0.0283987127	0	0
Pitau	Upper middle income	18.09199905	80.96799996	99.6574555	0	0.3424544984	0	99.65799903	0	0.2430149677	0	99.63420199	0	0.3657890078	0	0	99.6574555	99.6574555	0	0.3424544984	0	0
British Virgin Islands	High income	30.23699951	48.51499939	99.86438356	0	0.1356164384	0	99.86438356	0	0.1356164384	0	NAN	NAN	NAN	NAN	0	99.86438356	99.86438356	0	0.1356164384	0	0
Gibraltar	High income	33.69100189	100	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
San Marino	High income	33.93799973	97.49800055	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Liechtenstein	High income	38.13700104	14.41600037	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Saint Martin (French part)	High income	38.659	100	99.9927139	0	0.000728610507	0	99.9927139	0	0.000728610507	0	NAN	NAN	NAN	NAN	0	99.9927139	99.9927139	0	0.000728610507	0	0
Monaco	High income	39.24399948	100	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Faeroe Islands	High income	48.85001068	42.39799881	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
American Samoa	Upper middle income	55.1869996	97.15299992	99.73377166	0	0.2262283515	0	99.73377166	0	0.2262283515	0	NAN	NAN	NAN	NAN	0	99.73377166	99.73377166	0	0.2262283515	0	0
Greenland	High income	56.77199936	87.28000531	100.000017	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Northern Mariana Islands	High income	57.56699921	91.79799652	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Marshall Islands	Upper middle income	59.1400024	77.73999109	88.57249698	11.387979	0.0399740285	0	94.42972	5.39028	0.18	0	86.9	13.1	0	0	0	88.57249698	88.57249698	11.387979	0.0399740285	0	0
Bermuda	High income	62.7229998	100	99.90314002	0	0.9685998294	0	99.90314002	0	0.9685998294	0	NAN	NAN	NAN	NAN	0	99.90314002	99.90314002	0	0.9685998294	0	0
Andorra	High income	72.76499939	87.91600037	100.0000337	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Isle of Man	High income	85.03199768	52.88000282	99.075	0	0.925	0	99.075	0	0.925	0	NAN	NAN	NAN	NAN	0	99.075	99.075	0	0.925	0	0
United States Virgin Islands	High income	104.4229995	95.93000299	98.71826738	0	1.281732624	0	98.71826738	0	1.281732624	0	NAN	NAN	NAN	NAN	0	98.71826738	98.71826738	0	1.281732624	0	0
Tonga	Upper middle income	105.1999996	23.08699922	98.38149777	0	0.38149777	0	98.38149777	0	0.38149777	0	NAN	NAN	NAN	NAN	0	98.38149777	98.38149777	0	0.38149777	0	0
Kiribati	Lower middle income	119.459991	95.5039796	77.97890285	4.07899925	17.85217953	0	80.8941849	2.09689799	36.9620532	0.428717551	0	96.06869819	5.143545636	0.248601276	0	77.97890285	77.97890285	4.07899925	17.85217953	0	0
Guam	High income	168.7830048	94.93800354	99.6952	0	0.3048	0	99.6952	0	0.3048	0	NAN	NAN	NAN	NAN	0	99.6952	99.6952	0	0.3048	0	0
Saint Lucia	Upper middle income	183.6289978	18.8469996	96.8874457	1.81562166	1.28655029	0	96.8874457	1.73415225	1.45182918	0	97.20562919	2.166718886	0.6274589237	0	96.8874457	96.8874457	1.81562166	1.28655029	0	0	
Samoa	Lower middle income	186.4100037	17.8889994	91.83772455	6.52089952	1.41862348	0.2227520679	91.78032287	6.22073309	1.72789216	0.2712816327	92.10119	7.88881	0	0	0	91.83772455	91.83772455	6.52089952	1.41862348	0.2227520679	0
Sao Tome and Principe	NAN	288.1620008	85.81699992	93.78221635	0	6.21773652	0	93.78221635	0	6.21773652	0	NAN	NAN	NAN	NAN	0	93.78221635	93.78221635	0	6.21773652	0	0
French Polynesia	High income	280.8039917	61.97500229	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
New Caledonia	High income	285.499973	71.51799774	99.30520057	0	0.6946794278	0	99.30520057	0	0.6946794278	0	NAN	NAN	NAN	NAN	0	99.30520057	99.30520057	0	0.6946794278	0	0
Barbados	High income	397.3710032	31.18929996	98.51454542	0.2670005781	1.271794905	0	98.51454542	0.2670005781	1.271794905	0	NAN	NAN	NAN	NAN	0	98.51454542	98.51454542	0.2670005781	1.271794905	0	0
French Guiana	NAN	298.1620008	85.81699992	93.78221635	0	6.21773652	0	93.78221635	0	6.21773652	0	NAN	NAN	NAN	NAN	0	93.78221635	93.78221635	0	6.21773652	0	0
Vanuatu	Lower middle income	307.1499939	25.52001313	91.23119075	1.06238055	0	7.70642402	88.39702123	1.255132675	0	0	10.3478661	99.5	0.5	0	0	91.23119075	91.23119075	1.06238055	0	7.70642402	0
Iceland	High income	341.125	83.89795	99.9999721	0	0.00000279450015	0	99.9999721	0	0.00000279450015	0	NAN	NAN	NAN	NAN	0	99.9999721	99.9999721	0	0.00000279450015	0	0
Martinique	NAN	375.260148	88.1399939	99.84190708	0	0.158025018	0	99.84190708	0	0.158025018	0	NAN	NAN	NAN	NAN	0	99.84190708	99.84190708	0	0.158025018	0	0
Belize	Upper middle income	397.8210022	45.02001413	98.40196463	1.249110829	0.348917411	0	97.96502758	1.358317682	0.684747393	0	98.87899	1.12104	0	0	0	98.40196463	98.40196463	1.249110829	0.348917411	0	0
Guadeloupe	NAN	400.1270142	98.49899292	99.80312604	0	0.1988739613	0	99.80312604	0	0.1988739613	0	NAN	NAN	NAN	NAN	0	99.80312604	99.80312604	0	0.1988739613	0	0
Brunei Darussalam	High income	437.4830017	78.25000783	99.9000368	0	0.09996320036	0	99.9000368	0	0.09996320036	0	NAN	NAN	NAN	NAN	0	99.9000368	99.9000368	0	0.09996320036	0	0
Malta	High income	441.5390015	44.7440033	100.000004	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Maldives	Upper middle income	540.5419922	42.68699972	96.544386	0.04805095	0.40751076	0	96.544386	0.04805095	0.40751076	0	NAN	NAN	NAN	NAN	0	96.544386	96.544386	0.04805095	0.40751076	0	0
Caro Verde	Upper middle income	555.967971	66.65000433	88.76906452	7.904150846	0	0.1170950144	80.1452054	9.91172702	9.623217003	0.3511030597	93.1	6.9	0	0	0	88.76906452	88.76906452	7.904150846	0.1170950144	9.91172702	0
Suriname	Upper middle income	586.8339722	66.14900028	97.89863167	1.06724841	0.379336073	0.563784884	96.54741263	1.60295986	0.1598017947	1.66548972	98.7385387	0.794465231	0	0	0	97.89863167	97.89863167	1.06724841	0.379336073	1.06724841	0
Luxembourg	High income	625.9760132	91.45299549	99.9999324	0	0.1200674649	0	99.9999324	0	0.1200674649	0	NAN	NAN	NAN	NAN	0	99.9999324	99.9999324	0	0.1200674649	0	0
Montenegro	Upper middle income	628.0620117	97.48800098	98.8591652	0.545478804	0.552312011	0.00526050894	98.1618974	0	1.82133638	0.0162706977	99.19173699	0.808260428	0	0	0	98.8591652	98.8591652	0.545478804	0.552312011	0.00526050894	0
China, Macao SAR	High income	646.341168	100	100	0	0	0	100	0	0	0	NAN	NAN	NAN	NAN	0	100	100	0	0	0	0
Solomon Islands	Lower middle income	686.8779907	64.7000008	97.3102554	5.79547331	21.26784313	0.536503255	99.4052749	6.52832649	26.95591	7.11027957	91.47073859	3.557129425	3.899826137	1.132816513	0	97.3102554	97.3102554	5.79547331	21.26784313	0.536503255	0
Bhutan	Lower middle income	771.619995	42.3199988	97.3122263	2.460712607	0.17642813	0.0496045412	97.3193565	3.26800351	0	0.122178618	0	98.10662849	1.35914515	0.4169208475	0.1073061586	97.3122263	97.3122263	2.460712607	0.17642813	0.0496045412	0
Guyana	Upper middle income	786.559021	26.7699993	95.55486885	1.143089489	2.05244444	0.2068978897	93.5294905	1.644193682	1.561274833	2.86404089	100	0	0	0	0	95.55486885	95.55486885	1.143089489</			

Our findings

Investigating population size

Investigating global population size	
The total national population size (in 1000s)	7786095.168
The estimated world population (in 1000s)	7821000
Investigating urban population size	
The total urban population size (in 1000s)	4376308.463
The estimated urban population worldwide (in 1000s)	4361650
The calculated urban share (%)	56.15%

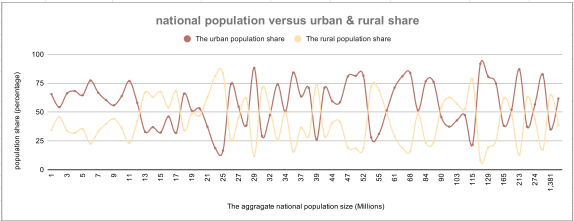
Total population in your dataset vs estimated world population (7.821 billion)	
The percentage difference (1)	0.436084719
The dataset population estimate is highly consistent with official global population figures. This indicates that our dataset covers almost the entire global population. A percentage difference of ~0.44% is very small.	
The dataset population estimate differs from the official 2020 world population by less than 0.5%, indicating that the dataset provides a nearly complete and globally representative coverage.	

Total population in your dataset vs estimated world population (7.821 billion)	
The percentage difference (2)	1.760119067
A percentage difference of ~1.7% is still low, but higher than for total population.	
This discrepancy arises because: This calculation is based on aggregated percentages. The other is based on absolute population counts.	
Aggregation of percentages across countries introduces weighting effects (large vs small countries).	

Overall conclusion

Both percentage differences are very small, confirming that the dataset is highly consistent with global population estimates. Minor discrepancies, particularly for urban population measures, are explained by aggregation effects and differences between absolute and percentage-based calculations rather than data quality issues.

When the urban share increases, the rural share decreases. This confirms the logical relationship: Urban share + Rural share = 100%. This validates that the data and transformations were done correctly.



By aggregating population into millions and averaging: - The chart becomes more readable. - Extreme outliers no longer dominate the x-axis. - Variability is still visible, which is analytically important. This confirms that the aggregation choice was appropriate.

As population size increases: - Urban and rural shares fluctuate significantly. - There is no clear upward or downward trend. - Countries with similar population sizes can have very different urbanization levels. Population size alone does not determine how urbanized a country is.

Across small, medium, and large population groups: - Some countries are highly urbanized. - Others remain predominantly rural. - This variability persists even among very large populations. Urbanization depends more on economic structure, development level, and policy, not just population size.

Overall conclusion

The visualization shows an inverse relationship between urban and rural population shares, as expected. However, no clear trend emerges between national population size and the degree of urbanization. Countries with similar population sizes can have very different urban-rural distributions, indicating that population size alone does not explain urbanization patterns.

Investigating access by area

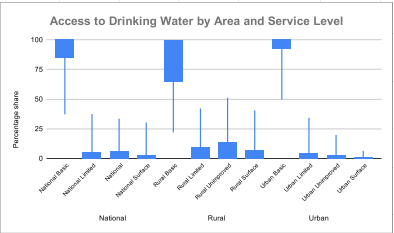
	wat_bas_n	wat_lim_n	wat_unimp_n	wat_sur_n	wat_bas_r	wat_lim_r	wat_unimp_r	wat_sur_r	wat_bas_u	wat_lim_u	wat_unimp_u	wat_sur_u
	National Basic	National Limited	National Unimproved	National Surface	Rural Basic	Rural Limited	Rural Unimproved	Rural Surface	Urban Basic	Urban Limited	Urban Unimproved	Urban Surface
MAX	100	37.42096287	33.53911377	30.36979308	100	42.16438068	51.21598167	40.51813242	100	34.27978009	19.83560681	6.362160532
MIN	37.20202005	0	0	0	21.98275234	0	0	0	49.66165495	0	0	0
MEAN	89.86476244	3.865265345	4.41801054	1.915488584	81.33591803	6.836562548	8.732261484	4.224039638	94.69893975	3.281838654	1.71878521	0.3131148443
MEDIAN	97.3481397	0.4707829181	0.8536474704	0	90.73161883	1.812330335	3.229749713	0.2196897375	98.10662849	0.5	0.3498740028	0
MODE	100	0	0	0	100	0	0	0	100	0	0	0
First quartile	85.6431096	0	0.0339692245	0	84.8294211	0	0.1584109149	0	92.6287712	0	0	0
Third quartile	99.88752524	4.851219157	5.279309922	1.878301993	99.11524249	8.553175126	13.2663393	6.157471481	99.94941803	3.966347184	2.059001302	0.1597056849
IQR	14.24421428	4.851219157	5.245403699	1.878301993	34.26382139	8.553175126	13.10792838	6.157471481	7.38654051	3.966347184	2.059001302	0.1597056849
Standard Deviation	15.0874269	6.96084633	7.174477036	3.922973987	21.50663741	8.704765846	11.82849434	6.847951395	8.03886161	5.6306705	3.25457032	0.8696316426

We observe that: - Mean & Median & Mode for all four features. - This tells us that: - The distributions are not normal (not symmetric). - This is expected in global data because: - The countries are very heterogeneous. - The access levels are clustered at extremes (near 0% or near 100%).

Interpretations

National share of population with at least basic drinking water access	wat_bas_n	Most countries have very high national access to basic drinking water, and many have full coverage. However, a small group of countries with very low access creates inequality at the global level. This is why the mean is lower than the median and why variability exists despite high overall access.
National share of population with limited drinking water service	wat_lim_n	In most countries, limited water access is very low or zero, which is why the median is close to 0 and the most common value is 0. However, a small group of countries still has significant limited access, raising the mean and increasing the spread.
National share of population with unimproved drinking water service	wat_unimp_n	Most countries have unimproved water access that is very low or near zero, which is why the median is small and the mode is 0. However, a smaller group of countries still has high unimproved access, pushing the mean upward and creating noticeable variability across the dataset.
National share of population using surface water for drinking water	wat_sur_n	Surface water use is zero for most countries, which is why the median and mode are 0 and the IQR is small. However, a few countries still rely heavily on surface water, which increases the maximum and makes the standard deviation larger than what the median alone suggests.

Rural share of population with at least basic drinking water access	wat_bas_r	Most countries have high rural basic water access and many are near 100%. However, a minority of countries have very low rural access, which creates strong inequality and explains the large IQR and standard deviation.
Rural share of population with limited drinking water service	wat_lim_r	Most countries have very low (often zero) rural limited access, but a minority of countries still has high limited service in rural areas, which raises the mean and increases variability.
Rural share of population with unimproved drinking water service	wat_unimp_r	Most countries have low rural unimproved access, but a minority of countries still has very high unimproved water use in rural areas. This is why the mean is higher than the median and why IQR/STD are relatively large.
Rural share of population using surface water for drinking water	wat_sur_r	Most countries have very low (often near zero) rural surface water use, but a few countries rely heavily on surface water in rural areas, which raises the maximum and increases variability.
Urban share of population with at least basic drinking water access	wat_bas_u	Urban basic water access is very high for most countries and often reaches 100%. The distribution is tight (small IQR and STD), meaning urban access is relatively similar across countries, with a few low-access exceptions.
Urban share of population with limited drinking water service	wat_lim_u	Urban limited access is usually very low or zero, but a few countries still show high limited service in urban areas, raising the maximum and increasing spread.
Urban share of population with unimproved drinking water service	wat_unimp_u	Urban unimproved access is near zero for most countries, but a few countries still have noticeable unimproved access, which increases the maximum and spread.
Urban share of population using surface water for drinking water	wat_sur_u	Urban surface water use is essentially zero for most countries (median/mode = 0, tiny IQR). Only a small number of countries have noticeable urban surface water reliance, which explains the maximum and the standard deviation.

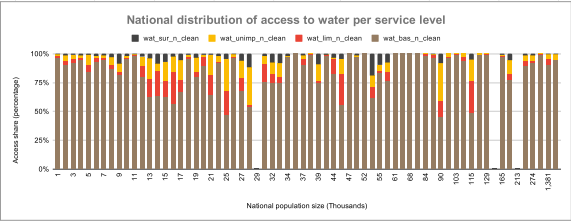


Interpretation by service type		
At least basic access	National (wat_bas_n)	box high (near the top) → most countries have high national basic access, but some lower outliers exist.
	Rural (wat_bas_r)	box lower and much wider → rural basic access varies a lot, some countries are far behind.
	Urban (wat_bas_u)	box highest and tightest → urban basic access is near universal in most countries.
Limited access	Urban (wat_lim_u)	box close to 0 → in most countries, urban limited access is almost zero.
	National (wat_lim_n)	also close to 0 but slightly higher spread than urban.
	Rural (wat_lim_r)	higher box and longer whiskers → a subset of countries still has meaningful rural limited access.
Unimproved access	Urban (wat_unimp_u)	box very close to 0 → unimproved access is rare in urban areas for most countries.
	National (wat_unimp_n)	low but slightly higher than urban.
	Rural (wat_unimp_r)	clearly higher box and long upper whisker → rural unimproved access is a major problem in some countries.
Surface water	Urban (wat_sur_u)	almost all values near 0 → urban surface water use is nearly nonexistent.
	National (wat_sur_n)	near 0 but with some countries higher.
	Rural (wat_sur_r)	higher and more spread → some countries rely heavily on surface water in rural areas.

Overall pattern across areas	
Urban is best and most consistent	
For wat_bas_u, the box is very high and relatively tight → most countries have very high urban basic access with limited variation.	
Rural shows the biggest inequality	
For wat_bas_r, the box is lower and much taller than urban → rural basic access is lower and much more unequal across countries.	
For "problem" indicators (wat_lim_r, wat_unimp_r, wat_sur_r), rural boxes and whiskers are generally higher → rural areas have more limited, unimproved, and surface water use.	
National sits between Urban and Rural	
National distributions usually fall between urban and rural, which makes sense because national values combine both.	

The box-and-whisker plots show clear differences by area. Urban basic water access is consistently high across countries, with a tight distribution close to 100%. Rural basic access is lower and much more variable, indicating stronger inequality between countries. For limited, unimproved, and surface water indicators, distributions are concentrated near zero in urban areas but are higher and more dispersed in rural areas, meaning these problems are primarily rural and concentrated in a subset of countries.

Investigating access by population size



Most population-size bins will be dominated by basic access (large blue section).
The remaining sections (limited/unimproved/surface) appear as smaller bands.
We don't see a strong "population size → access" pattern; differences are more influenced by development/income and rural/urban structure.